

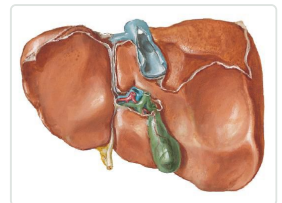
**BEFORE YOU READ — ANSWER FROM MEMORY**

1. What forms the portal vein, and where?
2. Name the boundaries of Calot's triangle and the artery found in it.
3. Which veins meet at each of the four porto-systemic anastomoses?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Liver — general features and position

What	Largest gland of the body · completely covered by a capsule, partially by peritoneum
Functions	Bile secretion · carbohydrate, fat and protein metabolism · heparin formation · waste removal · detoxification
Position	Right upper abdomen, under the right costal margin and right hemidiaphragm (which separates it from pleura, lung, pericardium, heart) · extends under the left hemidiaphragm
Surfaces	Antero-superior (diaphragmatic) and postero-inferior (visceral)
Lobes	Falciform ligament → right and left lobes · right lobe includes the caudate (C) and quadrate (Q) lobes
Functional lobes	Cantlie's line (principal plane, gallbladder → IVC) divides the liver into two functional lobes, each with portal segments

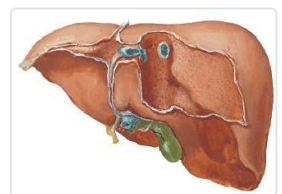


Visceral surface of the liver with the porta hepatis.

Relations	
Anterior	Diaphragm · right and left costal margins · right and left pleura · lower margins of both lungs · xiphoid · anterior abdominal wall in the subcostal angle
Posterior / inferior	Diaphragm and pleura · both lungs · right kidney · hepatic flexure · duodenum · gallbladder · IVC · esophagus · fundus of stomach

2 Porta hepatis, ligaments and supply

Porta hepatis	Hilum on the visceral surface, between caudate and quadrate lobes · lesser omentum attached to its margins
Contents	R and L hepatic ducts · R and L branches of hepatic artery · R and L branches of portal vein · sympathetic and parasympathetic nerves · hepatic lymph nodes
Peritoneal ligaments	Falciform · coronary (upper and lower layers) · triangular (right and left) · lesser omentum
Remnant ligaments	Ligamentum teres (obliterated left umbilical vein*) · ligamentum venosum (ductus venosus)
Bare area	Part of the posterior surface between the coronary layers with no peritoneum



Posterior view of the liver showing the bare area.

*The slide writes "Oblit. Lt. paraumb. v."; the ligamentum teres is usually described as the obliterated left umbilical vein.

Blood in	Hepatic artery ~30% · portal vein ~70%
Blood out	Hepatic veins (→ IVC)
Lymph	Liver makes about half of the body's lymph · porta hepatis nodes → celiac nodes · bare area → through the diaphragm → posterior mediastinal nodes
Nerves	Sympathetic and parasympathetic from the celiac plexus · anterior vagal trunk gives a hepatic branch

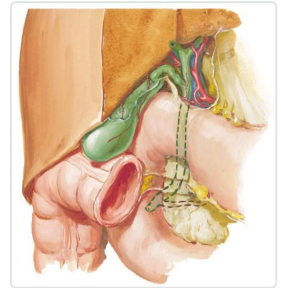
Clinical (headings on slides)	Liver support · liver trauma · liver biopsy · subphrenic spaces
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3 Extrahepatic biliary system

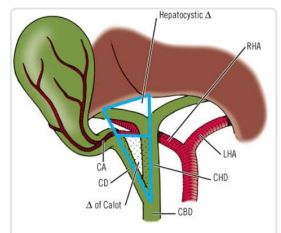
Part	Key facts
Hepatic ducts	Emerge from the porta hepatis · R + L unite → common hepatic duct (~4 cm) , which descends in the lesser omentum
Cystic duct	~ 3.8 cm , somewhat S-shaped · mucosa forms the spiral valve · joins the CHD → CBD
Common bile duct	~ 8 cm · 1st part in the lesser omentum · 2nd behind the duodenum (slide: "behind 2nd Duo.")* · 3rd behind the head of pancreas · joins the pancreatic duct → hepatopancreatic ampulla (of Vater) → major duodenal papilla , guarded by the sphincter of Oddi

*Textbooks place the second (retroduodenal) part behind the **first** part of the duodenum.

Gallbladder	Under the liver · stores 30-50 mL of bile · fundus, body, neck
Relations	Front: anterior abdominal wall, inferior surface of liver · behind: transverse colon, 1st and 2nd parts of duodenum
Supply	Cystic artery · cystic vein · a cystic lymph node at the neck · autonomic nerves from the celiac plexus · cholecystokinin (hormonal)
Calot's triangle	Bounded by the cystic duct, cystic artery and common hepatic duct
Hepatocystic triangle	Bounded by gallbladder wall + cystic duct, liver edge and CHD · the cystic artery (and so Calot's triangle) lies within it
Clinical (headings)	Gallstones · biliary colic (epigastric pain, jaundice, pancreatitis) · acute cholecystitis with pain referred to the shoulder · cholecystectomy (variations) · gangrene (single arterial supply) · biliary atresia



Extrahepatic biliary tree.

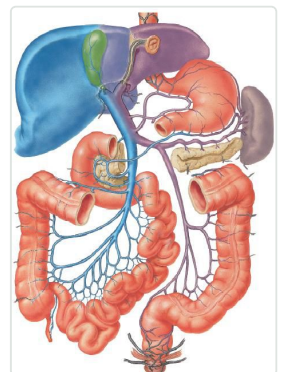


Calot's and hepatocystic triangles.

4 Portal vein

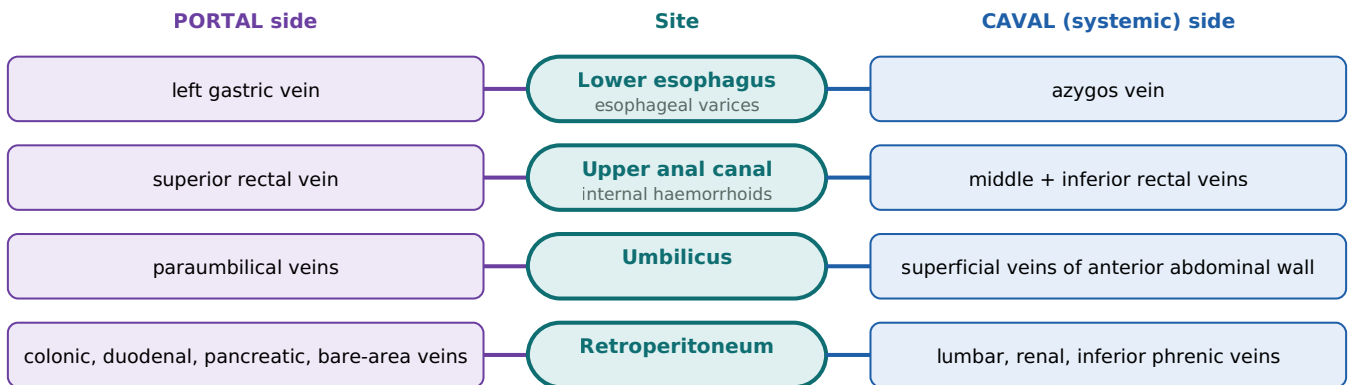
Length	About 2 in (5 cm)
Formed	Behind the neck of the pancreas by the SMV + splenic vein
Course	Up and right behind D1 → into the lesser omentum → in front of the epiploic foramen → porta hepatis → right and left branches
Drains	Abdominal GIT · lower 1/3 of esophagus · upper half of anal canal* · spleen, pancreas, gallbladder
Tributaries	Splenic, superior mesenteric, left gastric , right gastric, cystic veins
Clinical (headings)	Portal hypertension · portocaval shunt · liver secondaries via the portal vein

*LG 3 (Large Intestine) gave the portal territory of the anal canal as the upper two-thirds, above the pectinate line.



The portal vein and its tributaries.

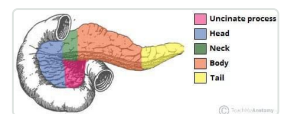
PORTO-SYSTEMIC ANASTOMOSES — where portal blood can bypass the liver



In portal hypertension blood is forced through these channels and the veins dilate (varices, haemorrhoids).

5 Pancreas

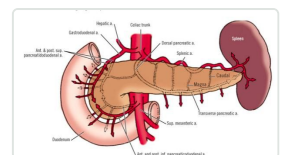
Function	Exocrine (acinar enzymes, duct bicarbonate) + endocrine islets: β cells → insulin , α cells → glucagon
Parts	Head (lower part projects as the uncinate process) · neck · body · tail
Position	Retroperitoneal, across the posterior wall around the transpyloric plane



Parts of the pancreas.

Part	Relations
Head	In the C of the duodenum : D1 above, D2 on the right (with the superior and inferior pancreaticoduodenal arteries between), D3 below · front: transverse colon · behind: IVC, renal veins, common bile duct
Uncinate process	Between the abdominal aorta (behind) and the superior mesenteric vessels (in front)
Neck	Front: gastroduodenal junction · behind: formation of the portal vein (SMV + splenic vein)
Body — surfaces	Anterior : stomach, separated by the lesser sac · inferior : DJ flexure, loops of jejunum, end of transverse colon · posterior : aorta + SMA origin, splenic and left renal veins , left psoas, left crus, left kidney, left suprarenal, left sympathetic chain
Body — borders	Superior : splenic artery · anterior : transverse mesocolon and greater omentum attach · inferior : separates inferior from posterior surface
Tail	Reaches the hilum of the spleen through the lienorenal (splenorenal) ligament

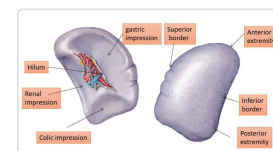
Duct	Drains	Opens
Main	Upper head, body, tail · runs tail → head	Joins CBD → ampulla of Vater → major papilla, D2 posteromedial
Accessory	Uncinate process + lower head	D2 above the ampulla
Arteries (slide figure)	Superior pancreaticoduodenal (gastroduodenal) · inferior pancreaticoduodenal (SMA) · splenic branches: dorsal pancreatic, pancreatica magna, caudal, transverse pancreatic	
Lymph	Upper head → celiac · lower head → superior mesenteric · rest → pancreaticosplenic nodes	



Arterial supply of the pancreas.

6 Spleen

Odd numbers 1-3-5-7-9-11	1 × 3 × 5 in (2.5 × 7.5 × 12.5 cm) · 7 oz (~220 g) · lies along left ribs 9-11
Surfaces	Diaphragmatic (diaphragm) · visceral : gastric, renal and colic impressions + hilum
Borders · ends	Superior and inferior borders · anterior and posterior extremities
Visceral relations	Front: fundus of stomach · behind: left suprarenal and kidney · below: tail of pancreas and left colic flexure
Splenic artery	Very tortuous · enters the hilum between layers of the splenorenal ligament · 5-6 branches into the spleen
Splenic vein	Usually behind the artery, less tortuous · receives the inferior mesenteric vein · joins the SMV to form the portal vein
Lymph	Hilar nodes → along splenic vessels → pancreaticosplenic nodes



Surfaces and impressions of the spleen.

P Must-remember pearls

- Liver: largest gland · hepatic artery 30% / portal vein 70% · drains via hepatic veins to IVC · makes ~½ of body lymph.
- Porta hepatis lies between caudate and quadrate lobes: hepatic ducts, hepatic artery branches, portal vein branches, nerves, nodes.
- Ligamentum teres = obliterated umbilical vein · ligamentum venosum = ductus venosus · bare area has no peritoneum and drains to posterior mediastinal nodes.
- CHD ~4 cm · cystic duct ~3.8 cm with spiral valve · CBD ~8 cm: lesser omentum → behind duodenum → behind pancreatic head → ampulla of Vater.
- Calot's triangle: cystic duct, cystic artery, CHD · gallbladder gangrene risk from its **single** cystic artery.
- Portal vein: 5 cm, **SMV + splenic behind the pancreatic neck**, in front of the epiploic foramen.
- Porto-systemic: left gastric ↔ azygos · superior rectal ↔ middle/inferior rectal · paraumbilical ↔ abdominal wall veins · retroperitoneal ↔ lumbar/renal/phrenic.
- Pancreas: head in duodenal C (CBD behind) · uncinate between aorta and SMV/SMA · neck over portal vein formation · tail in the splenorenal ligament.
- Main pancreatic duct + CBD → ampulla of Vater; accessory duct (uncinate, lower head) opens above it.
- Spleen 1-3-5-7-9-11 · artery tortuous, in the splenorenal ligament · vein receives the IMV.

Q Test yourself

1. During a Whipple procedure the surgeon lifts the neck of the pancreas. Which structure is formed directly behind it?

- A. Common bile duct
- B. Portal vein
- C. Inferior vena cava
- D. Splenic artery
- E. Left renal vein

2. During cholecystectomy the surgeon dissects the triangle bounded by the cystic duct, common hepatic duct and cystic artery. Which triangle is this?

- A. Hepatocystic triangle
- B. Lumbar triangle
- C. Calot's triangle
- D. Hesselbach's triangle
- E. Submandibular triangle

3. A man with cirrhosis vomits blood from dilated veins in his lower esophagus. Which portal tributary feeds them?

- A. Superior rectal vein
- B. Paraumbilical vein
- C. Inferior mesenteric vein
- D. Left gastric vein
- E. Splenic vein

4. A carcinoma in the head of the pancreas causes obstructive jaundice. Which posterior relation of the head explains this?

- A. Common bile duct
- B. Splenic artery
- C. Left renal vein
- D. Superior mesenteric artery
- E. Left crus of diaphragm

5. A footballer with an enlarged spleen is kicked on the left side. Between which ribs does the spleen normally lie?

- A. 5th–7th
- B. 7th–9th
- C. 11th–12th
- D. 1st–3rd
- E. 9th–11th

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 B — The portal vein is formed behind the neck of the pancreas by the SMV and splenic vein.

2 C — Calot's triangle is bounded by the cystic duct, cystic artery and common hepatic duct; the hepatocystic triangle uses the liver edge instead of the artery.

3 D — At the lower esophagus the left gastric vein (portal) anastomoses with the azygos vein (systemic).

4 A — The common bile duct runs behind the head of the pancreas, so a tumour there compresses it.

5 E — By the rule of odd numbers the spleen lies along the left 9th to 11th ribs.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Anatomy GIT Block · MEDucation by Mattin Majid · GIT Anatomy SGL 4 slides, 2026-2027
Study alongside the lecture, not instead of it.